Tyler Wyatt

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Education

George Mason University — **Fairfax, VA**
B.S. in Computer Science (GPA 3.84) — Aug. 2021 – May 2024
M.S. in Computer Science (GPA 3.72) — Aug. 2024 – May 2025

Experience

Graduate Research Assistant — 32 Hours/Week, Oct. 2024 – Present
Wireless, IoT, and Sensing Lab, George Mason University — *Fairfax, VA*

- Independently identify research gaps in ML-based propagation modeling and develop viable project proposals through regular meetings with PIs
- Develop surrogate models for wireless raytracing simulators using state-of-the-art transformer-based mesh neural rendering techniques for accelerating power delay spectrum estimation
- Architect distributed ETL pipelines to process 3D urban meshes from OpenStreetMap and generate 2 million simulated propagation environments for training using HPC resources
- Benchmark model performance against CNN, GNN, and empirical baselines to quantify improvements in prediction accuracy and inference speed

Device Engineering Intern — 40 Hours/Week, Jun. 2024 – Aug. 2024
Embedded Linux Team, Alarm.com — *Tysons, VA*

- Migrated Alarm Hub service from an MCU to a Telit LE910C4-NF module, eliminating the MCU requirement to reduce manufacturing costs and product complexity while maintaining feature parity with existing hub products
- Planned and implemented the hardware abstraction layer to route existing Z-Wave and AT command subsystems communication via shared memory and UART devices
- Automated the management of routing, firewalls, and kernel drivers across Wi-Fi, Ethernet, and cellular interfaces to ensure high-availability for connected IP camera video streams and smart home appliances
- Performed RF stress testing using an Amari Callbox and iperf3 to verify network stability under high network and CPU load

Assistant Instructor — 12 Hours/Week, Mar. 2020 – Oct. 2024
Elite M.A. Centers — *Springfield, VA*

- Planned and taught class content for students of all ages
- Led daily summer camp activities

Projects

Incremental Process Checkpointing | C, Linux, KVM/QEMU — Spring 2025

- Implemented custom Linux system calls to checkpoint process memory by traversing virtual memory areas and dumping memory ranges to files
- Modified the kernel page fault handler to track modified pages on first writes, enabling incremental checkpointing with minimal runtime overhead
- Designed and built a custom filesystem to store process dumps

Personal Homelab | Linux, WireGuard, Proxmox VE

- Manage virtual machines using Proxmox VE hypervisor for network services, shared storage, and persistent compute tasks
- Configured a OpenMediaVault VM with SATA controller passthrough to manage network storage shares and automated daily backups
- Deployed a WireGuard tunnel to provide external access to shared storage and compute for multiple users

Technical Skills

Languages: Python, C, Java, Bash, LaTeX
ML/Data Science: PyTorch, PyTorch Geometric, NumPy, Matplotlib, ETL Pipelines
Systems: Linux Kernel Development, Embedded Linux, Networking
Wireless/RF: Sionna-RT, Propagation Modeling, LTE/Cellular, RF Testing
Tools/Infrastructure: Git, HPC (SLURM), Docker, KVM/QEMU, WireGuard, GNU Parallel